Findings

This study tests whether macro-regime conditioning improves the extraction of predictive signals from noisy macroeconomic features.

- **Target Signal:** Macro-regime conditioning provides significant risk-adjusted outperformance in commodity-sensitive equities (XOM, BA.L).
- **Model Parsimony:** A 2-state HMM structure demonstrates superior out-of-sample robustness compared to higher-order models.
- **Risk Mitigation:** The framework functions primarily as a selective downside filter (tail-risk mitigation) rather than a continuous alpha generator.

Inter-model Comparison

To assess model parsimony, three Elastic Net specifications were evaluated:

- Baseline: a regime-unaware baseline (ElasticNet before HMM analysis)
- elasticNetHmmProb: Hmm regime features are added to training
- elasticNetRegimeSpec: The original training matrix is altered by the probabilities of the regimes.

The Exxon stock, XOM, was used as the informative stock due to its known sensitivity to macro and commodity-driven factors. Figure 1 shows clear separation between baseline and regime-aware models across all feature sets. Table 1 shows that, beyond Sharpe ratios, the baseline model also performed worse in both drawdown and had a lower percentage of hit rate. Expanding the analysis to post HMM shows that the drawdown and hit rate perform better in the probability informed elastic net, this is consistent across features. Table 1 also shows consistently large drawdowns (>50%) and hit rates close to 50%, indicating the model does not derive value from directional accuracy. Instead, performance arises from selective exposure in volatile environments and avoiding adverse conditions.

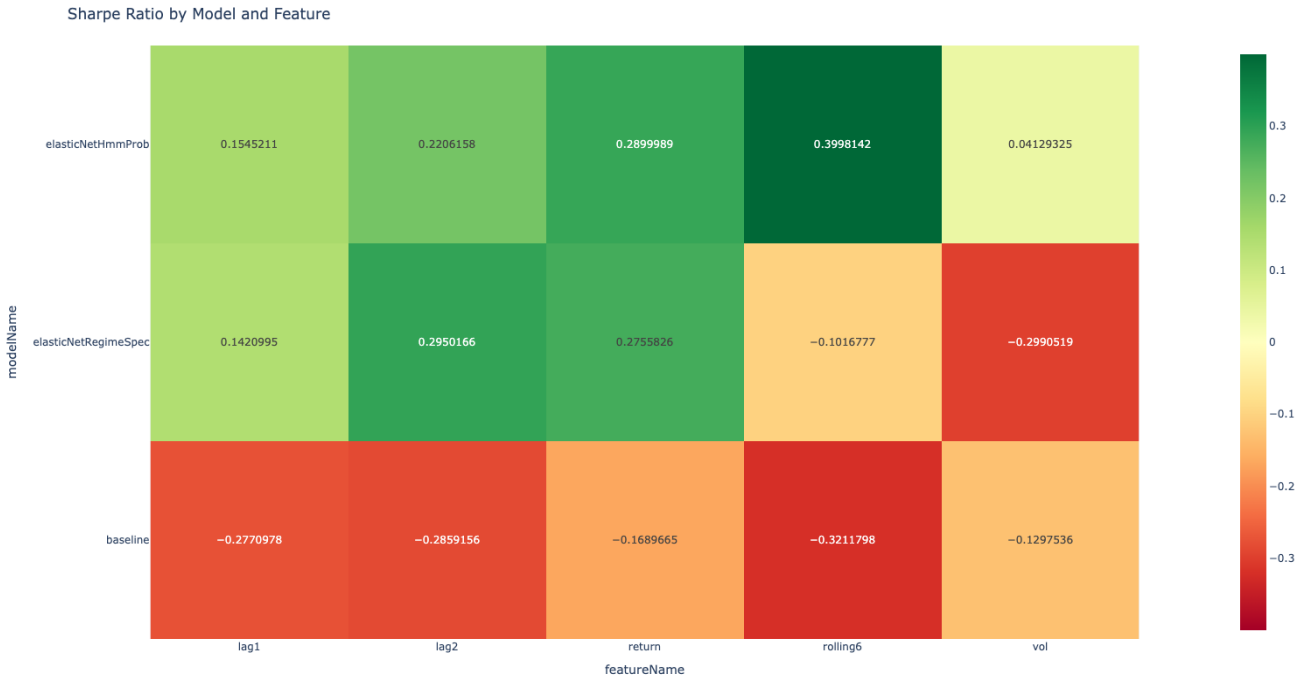

Figure 1: Sharpe ratios by model and feature set of XOM stock with a 0.7 threshold.

The model was able to beat buy and hold strategies for monetary returns when analysing the return feature set as well as in the feature lagged by two months. Furthermore, the regime feature aware elastic net, elasticNetHmmProb, shows greater Sharpe ratio than the regime probability adjusted elastic net, elasticNetRegimeSpec, showing that inclusion of the regime within elastic nets enables capturing of the weak macro signal.

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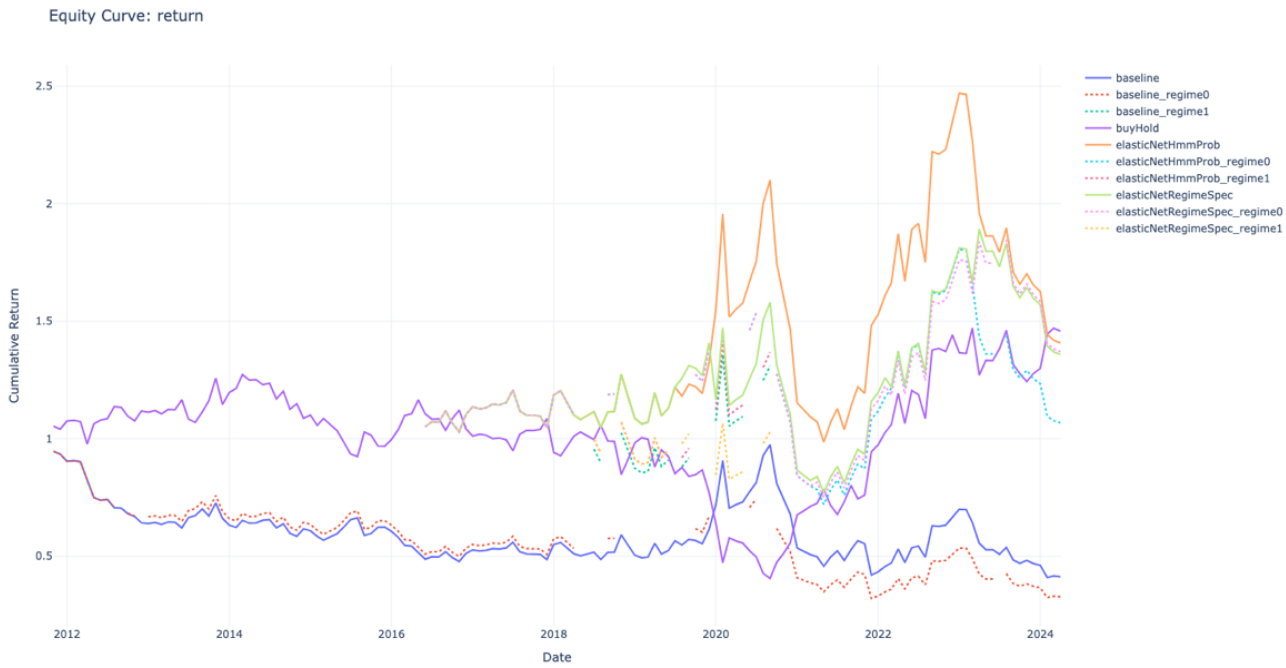

Figure 2: Equity curves across all models and regimes of XOM stock with a 0.7 threshold.

	Lag1	Lag2	Return	Rolling6	Volatility
Total Return	0.02554	0.1912	0.407	0.7389	-0.2169
Buy Hold return	0.3183	0.3457	0.2509	0.2991	0.2991
Sharpe ratio	0.1545	0.2206	0.29	0.3998	0.0413
Hit Rate	0.5213	0.5269	0.5474	0.5169	0.5281
Max drawdown	-0.6813	-0.6557	-0.5311	-0.3242	-0.6447

Table 1: Table of validation metrics from XOM stock with a 0.7 threshold

For simplicity all further analysis was according to results from ElasticNetHmmProb, as the model consistently showed positive Sharpe values across features, and the 'Return' feature, as the representative feature set due to its consistent performance across models and clear separation between regime-aware and baseline approaches.

Regime complexity reduces sensitivity

To investigate the impact of regime number on signal extraction, XOM was run on two and three regimes. The two-regime model displays relatively high Sharpe values in the regime aware models compared to the three-regime alternative. The two-regime model manages to reach Sharpe ratios greater than 0.1 in all but the rolling 6 month mean window (elasticNetRegimeSpec) and the volatility feature, whereas the three-regime feature, figure 3, only reaches above 0.1 Sharpe in 5/10 of regime aware models (7/10 in 2 regime). The rolling6 is a measure of mean over a six-month period and therefore dampens an already weak signal. Volatility is endogenised into the model as the variance in Sharpe Ratio between models is reduced in the two-regime pipeline in comparison to the three. Further to this, the endogenisation of this signal appears to assist in reducing the noise comparatively in the remaining engineered feature sets, with all features showing relatively higher differences between models compared with the three-regime system.

Figure 3: Sharpe ratios by model and feature set of XOM stock with a 0.7 threshold and 3 regimes.

Validation of efficacy through macro-insensitive stocks

To ensure that models had not been overfitted and results were determined by underlying extracted signals, a stock suspected to be less sensitive to macro influences was chosen, AAPL. The pipeline failed to delineate any signal that performed better than buy and hold strategies, as seen in Table 2. Furthermore, all models within the pipelines failed to report positive Sharpe ratios regardless of feature or model, implying that the previous improvements seen in the hmm aware elastic nets, were due to the Hmm itself and not an artifact of model overfitting or iteration. This was further shown with the use of further macro-insensitive stocks such as GOOG.

	XOM	AAPL	GOOG
Total Return	0.407	-0.795	-0.8631
Buy Hold return	0.2509	7.0439	4.027
Sharpe ratio	0.29	-0.5844	-0.9113
Hit Rate	0.5474	0.4316	0.4316
Max drawdown	-0.5311	-0.775	-0.848

Table 2: Validation metrics for XOM, AAPL, and GOOG with a 0.7 threshold.

Alpha signal capture is not uniform across macro sensitive stocks

To ensure that the pipeline's capabilities of alpha extraction in macro sensitive stock, another macro sensitive stock was chosen, BA.L. Encouragingly, this stock showed even better Sharpe values compared with XOM, with all engineered features displaying improved Sharpe ratios with regime awareness (Figure 4). In spite of this, the buy hold strategy beats the model's return value by over 6x. This is consistent with another macro-sensitive stock, CVX. Interestingly, as seen in figure 5, CVX displays poor Sharpe values across the model. In both stock cases, the equity curves show that the model is unable to recover the upward trend shown by the buy hold strategy, preventing the model from being able to capitalise on the resultant higher returns.

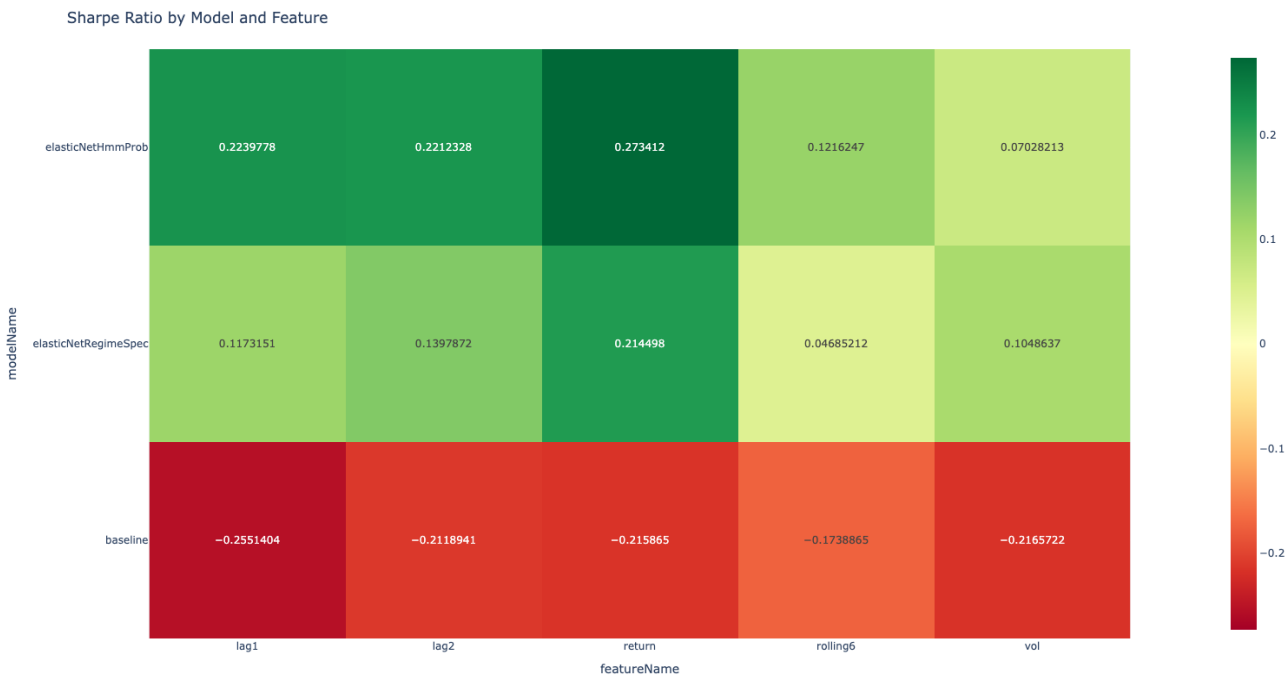

Figure 4: Sharpe ratios by model and feature set of BA.L stock with a 0.7 threshold.

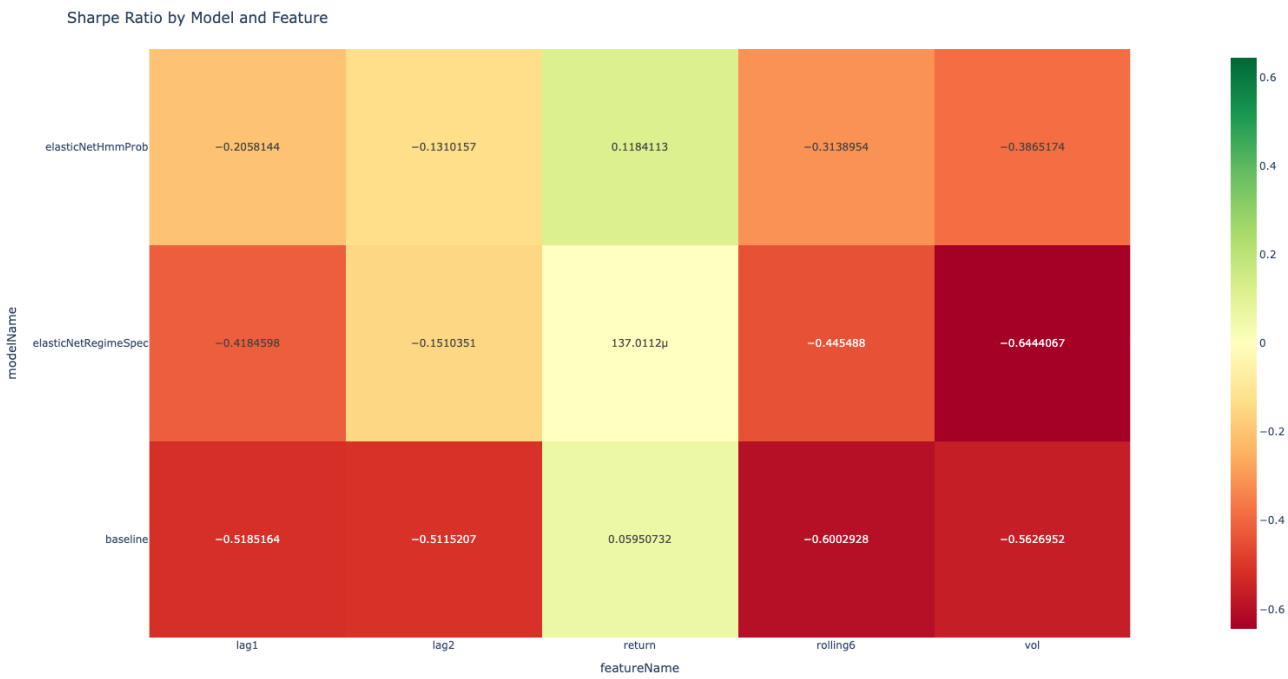

Figure 5: Sharpe ratios by model and feature set of CVX stock with a 0.7 threshold.

The model acts as a selective filter, not a consistent alpha extractor

To investigate the signal relationship further, it was expected that reducing the threshold for signal switching (previously 0.7), would enable the model to capitalise on the high returns seen in buy hold strategies by enabling more frequent trades. Indeed, reducing the signal threshold from 0.7 to 0.3 increased trading frequency but degraded both Sharpe ratio and total return across macro-sensitive assets, demonstrating that predictive power is concentrated in high-confidence signals, while weaker signals introduce noise. This is exemplified as the equity curve in figure 6 (threshold 0.3), approaches the

buy hold equity curve (although fails to exceed it), while the higher threshold figure 7, the equity curve remains much lower.

	XOM			BA.L			CVX		
	Threshold 0.3	Threshold 0.7	Difference	Threshold 0.3	Threshold 0.7	Difference	Threshold 0.3	Threshold 0.7	Difference
Total Return	0.2875	0.407	-0.1195	-0.1653	0.2914	-0.4568	0.1642	-0.0535	0.2177
Buy Hold return	0.2509	0.2509		1.6565	1.6565		0.5482	0.5482	
Sharpe ratio	0.2514	0.29	-0.0386	0.0313	0.2734	-0.2421	0.211	0.1184	0.0926
Hit Rate	0.5477	0.5474		0.5579	0.5579		0.5053	0.5053	
Max drawdown	-0.5529	-0.5311	-0.0219	-0.5651	-0.3097	-0.2554	-0.4352	-0.4727	0.0376

Table 3: Validation metrics for threshold of 0.7 and 0.3, and the differences between them, for XOM, BA.L, and CVX.

The third macro-sensitive stock, CVX, carries the inverse trend, with Sharpe ratio and Total Return increasing with a lower threshold. This is shown to be due to the reduced threshold allowing for greater capture of returns, approaching but not exceeding those seen in buy hold strategies. Whilst the increase trading allows for greater capture, the nature of the model as a risk mitigator prevents the exceeding of buy hold strategies (Figure 6, 7).

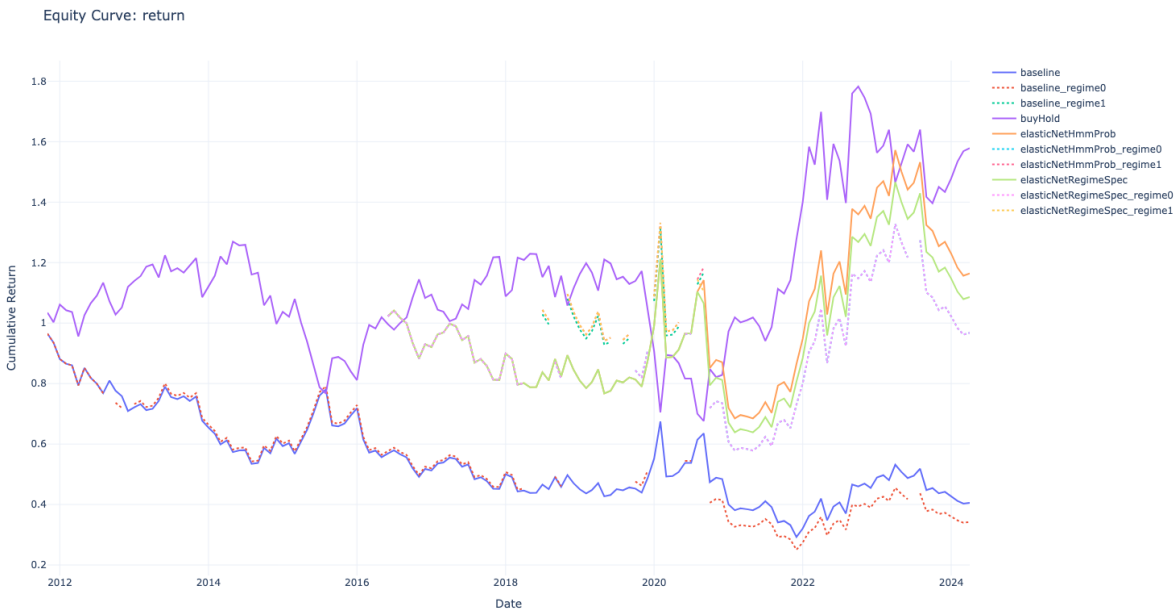

Figure 6: Equity curve returns for CVX per model, per regime with a threshold of 0.3.

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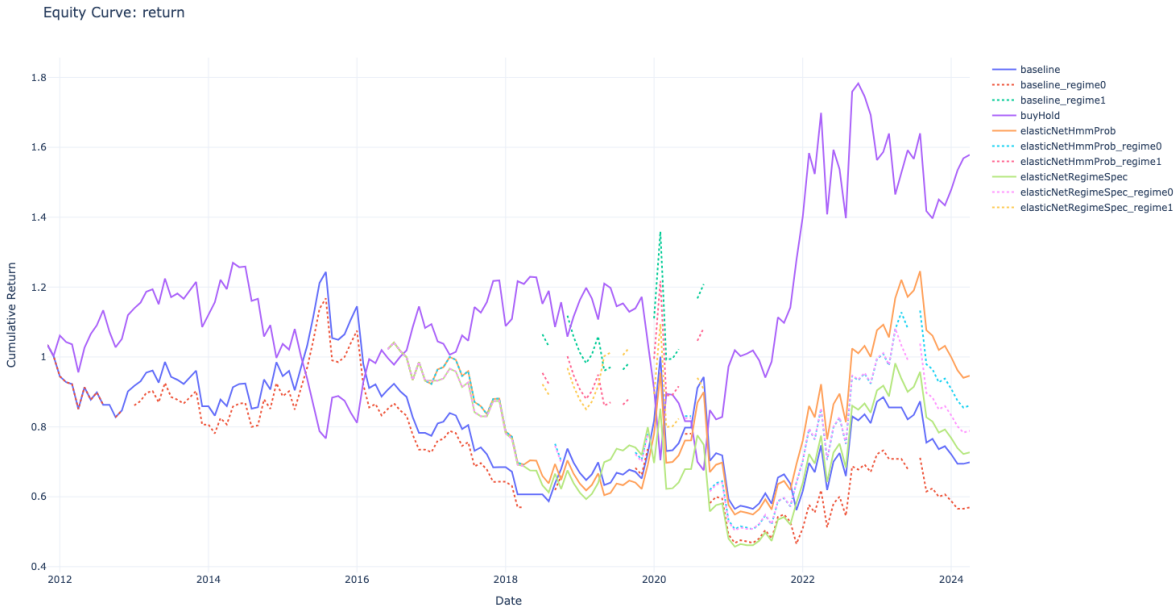

Figure 7: Equity curve returns for CVX per model, per regime with a threshold of 0.7.

Conclusions

Model Behaviour:

- The model is a tail risk mitigator, filtering when to trade rather than capturing continuous alpha
- Performance improvements arise from avoiding adverse conditions rather than capturing full upside trends.

Characteristics of signals:

- The signals captured are weak and therefore sensitive to model parameters (regimes, thresholds etc.)
- Consistently large drawdowns and hit rates close to 50% indicate that the model does not possess strong directional predictive power. Instead, performance improvements arise from selective exposure and the ability to mitigate adverse periods within noisy, volatile environments.

Applicability and Limitations:

- The model is only able to capture signals from macro-sensitive stock
- Stocks within macro-sensitive sectors still contain nuances that affect the ability of the framework to capture any signal

The model acts as a selective filter for tail risk mitigation, more so than an extractor of continuous alpha signals. The decrease in Sharpe values and total return seen in the macro-sensitive stocks, XOM and BA.L, when thresholds are lowered shows that an increase in trading frequency lowered the efficacy of the framework. This highlights its suitability for tail risk mitigation. Interestingly, the trend is not captured in the third macro-sensitive stock CVX where performance increases with the lowered threshold however the model still falls below buy and hold. This is seen as, although the lower threshold does allow a closer capture of the signal seen in the equity curves of buy and hold, it still mitigates and fails to capture the entire alpha signal. This demonstrates its efficacy as a tail risk mitigator. These findings indicate that macro signals provide conditional, not universal, predictive power, with effectiveness determined by regime structure, signal strength, and asset-specific dynamics.